Numerical Solutions to Systems of Linear Equations

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Introduction to system of Linear Equation

Systems of Equations

A set of equations is called a **system of equations**.

The **solutions** must satisfy each equation in the system.

If all equations in a system are linear, the system is a **system of linear equations**, or a **linear system**.

- a linear equation in n variables:

$$a_1x_1+a_2x_2+a_3x_3+\cdots+a_nx_n=b$$

$a_1, a_2, a_3, \ldots, a_n$, b: real number

a_1: leading coefficient

x_1: leading variable

Systems of Linear Equations:

A solution to a system of equations is an ***ordered pair*** that satisfy all the equations in the system.

A system of linear equations can have:

1. Exactly one solution
2. No solutions
3. Infinitely many solutions

Systems of Linear Equations:

There are four ways to **solve systems of linear equations**:

1. By graphing
2. By substitution
3. By elimination
4. By multiplication (Matrices)

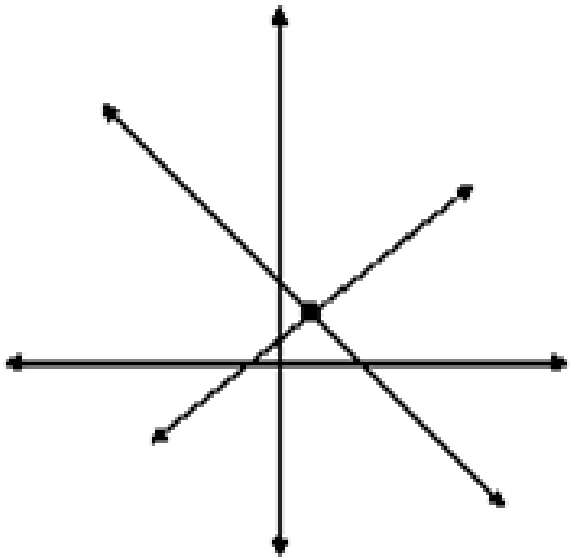
Solving Systems by Graphing:

When solving a system by graphing:

1. Find ordered pairs that satisfy each of the equations.
2. Plot the ordered pairs and sketch the graphs of both equations on the same axis.
3. The coordinates of the point or points of intersection of the graphs are the solution or solutions to the system of equations.

Solving Systems by Graphing:

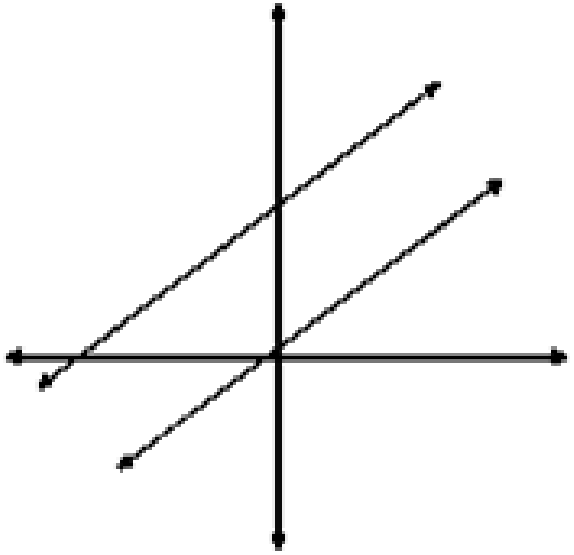
Consistent



One solution

Lines intersect

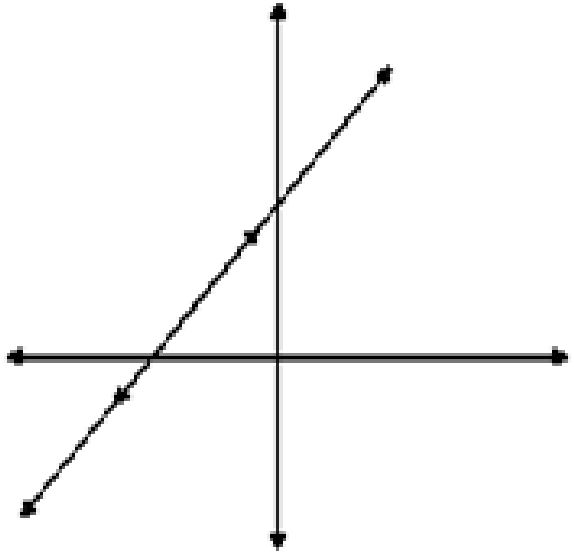
Inconsistent



No solution

Lines are parallel

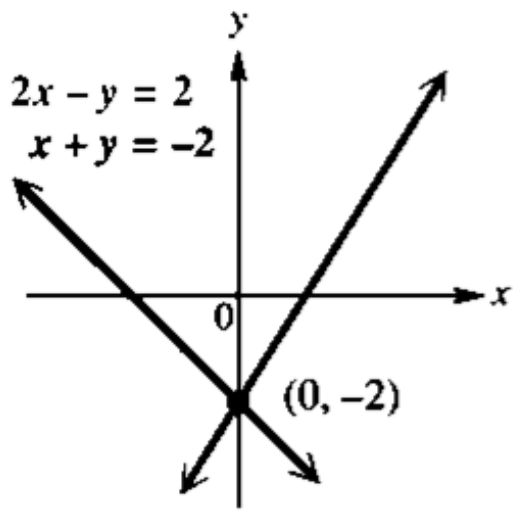
Dependent



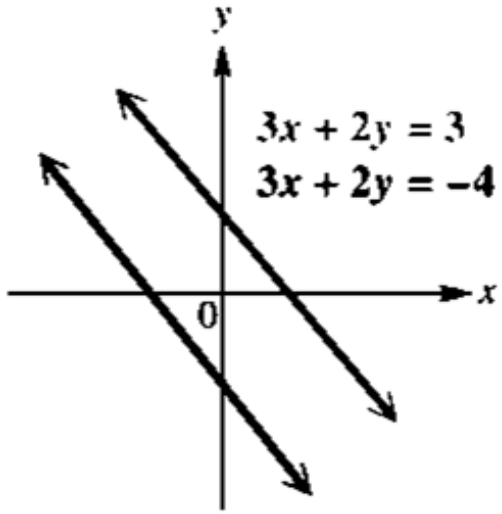
Infinite number of solutions

Coincide-Same line

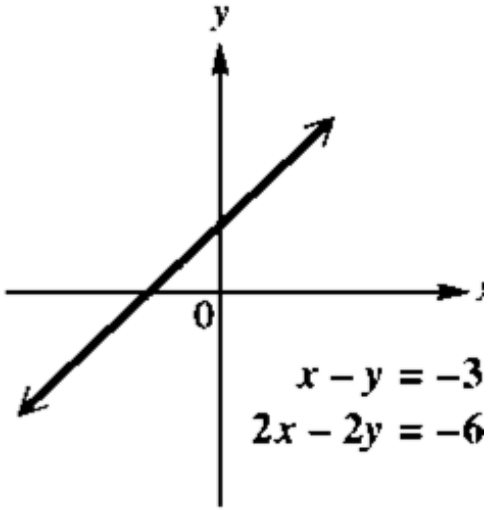
Linear System in Two Variables



One Solution



No Solutions



Infinitely Many Solutions

Three possible solutions to a linear system in two variables:

One solution: coordinates of a point

No solutions: **inconsistent** case

Infinitely many solutions: **dependent** case

Determine Without Graphing:

- There is a somewhat shortened way to determine *what type* (one solution, no solutions, infinitely many solutions) of solution exists within a system.
- Notice we are <u>not</u> finding *the* solution, just *what type* of solution.
- Write the equations in slope-intercept form: $y = mx + b$.

  (i.e., solve the equations for y, remember that *m = slope, b = y - intercept*).

Once the equations are in slope-intercept form, compare the slopes and intercepts.

One solution – the lines will have different slopes.

No solution – the lines will have the same slope, but different intercepts.

Infinitely many solutions – the lines will have the same slope and the same intercept.

Given the following lines, determine what type of solution exists, **without graphing**.

Equation 1: $3x = 6y + 5$

Equation 2: $y = (1/2)x - 3$

Writing each in slope-intercept form (solve for y)

Equation 1: $y = (1/2)x - 5/6$

Equation 2: $y = (1/2)x - 3$

Since the lines have the same slope but different y-intercepts, there is no solution to the system of equations. The lines are parallel.

2. Method of Substitution

Solving Linear Systems of two variables by Method of Substitution.

Example 6: Solve the system : $4x + y = 5$

$2x - 3y = 13$

Step 1: Choose the variable y to solve for in the top equation:

$$y = -4x + 5$$

Step 2: Substitute this variable into the bottom equation

$$2x - 3(-4x + 5) = 13 \longrightarrow 2x + 12x - 15 = 13$$

Step 3: Solve the equation formed in step 2

$$14x = 28 \qquad x = 2$$

Step 4: Substitute the result of Step 3 into either of the original equations and solve for the other value. $4(2) + y = 5$

$$y = -3$$

Solution Set: {(2,-3)}

Step 5: Check the solution and write the solution set.

Substitution Method:

1. Solve the following system of equations by substitution.

$$y = x + 3$$
$$x + y = -5$$

Step 1 is already completed.

Step 2:Substitute x+3 into 2nd equation and solve.

$$x + (x + 3) = -5$$
$$2x + 3 = -5$$
$$2x = -8$$
$$x = -4$$

Step 3: Substitute –4 into 1st equation and solve.

$$y = x + 3$$
$$y = -4 + 3$$
$$y = -1$$

The answer: (-4 , -1)

Substitution Method

Example Solve the system.

$$3x + 2y = 11 \quad \textbf{(1)}$$
$$-x + y = 3 \quad \textbf{(2)}$$

Solution

$y = x + 3$ Solve **(2)** for y.

$3x + 2(x + 3) = 11$ Substitute $y = x + 3$ in **(1)**.

$3x + 2x + 6 = 11$ Solve for x.

$5x = 5$

$x = 1$

$y = 1 + 3$ Substitute $x = 1$ in $y = x + 3$.

$y = 4$ Solution set: $\{(1, 4)\}$

Your Turn:

$$3x - y = 4$$

$$x = 4y - 17$$

Your Turn:

$$2x + 4y = 4$$
$$3x + 2y = 22$$